

DATA PREPROCESSING REPORT

INTRODUCTION.

The purpose of this data preprocessing stage was to transform the Telco Customer Churn dataset into a clean and machine-learning-ready format. The preprocessing involved data inspection, cleaning, feature engineering, visualization, outlier detection, feature selection, categorical encoding, and feature scaling.

The original dataset contained 7,043 customer records and 21 columns.

DATA INSPECTION.

The dataset was first inspected to understand its structure, data types, dimensions, and statistical characteristics.

The original dataset contained:

7,043 rows

21 columns

Numerical and categorical variables

Churn as the target variable

The numerical variables included SeniorCitizen, tenure, MonthlyCharges, and TotalCharges, while several customer and service-related variables were categorical.

MISSING-VALUE TREATMENT.

Missing values were investigated during the data-cleaning stage.

The TotalCharges column contained blank entries. These were handled during dataset import by treating blank spaces as missing values using:

```
na_values=("")
```

The affected records corresponded to customers with zero tenure, meaning they were new customers who had not accumulated any total charges.

After the appropriate cleaning and conversion of TotalCharges, the dataset was checked again to ensure that missing values were properly handled.

The final processed dataset contained no missing values.

DUPLICATE-VALUE CHECK.

Duplicate records were checked to ensure that the same customer record was not unnecessarily repeated.

The final processed dataset contained:

0 duplicate rows

Therefore, no duplicate records required removal.

The customerID column was also removed because it functions as an identifier rather than a meaningful predictive feature.

FEATURE ENGINEERING.

Feature engineering was performed to create an additional feature that summarizes the number of services used by each customer.

TotalServices

The feature was created by combining relevant service-related variables.

The purpose was to provide a single numerical representation of the number of services associated with each customer.

This feature was evaluated during feature selection.

An additional feature, AverageMonthlySpend, was also initially created during the feature-engineering process. However, correlation analysis showed that it had an extremely strong correlation of approximately 0.99 with MonthlyCharges.

Because this indicated substantial redundancy, AverageMonthlySpend was removed.

This demonstrates that feature engineering does not mean every engineered feature must be retained; features should be evaluated for usefulness and redundancy.

DATA VISUALIZATION.

Data visualization was used to better understand the distributions and relationships within the dataset.

The visualizations included:

Customer churn distribution

Customer tenure distribution

Monthly charges distribution

Customer distribution by contract type

Customer distribution by internet service

Correlation heatmap

Tenure versus total charges scatter plot

These visualizations helped identify patterns, distributions, and relationships among the variables before final preprocessing.

OUTLIER DETECTION.

Potential outliers were investigated using the Interquartile Range (IQR) method.

The numerical variables examined included:

tenure

MonthlyCharges

TotalCharges

AverageMonthlySpend during the stage when it existed

The analysis identified zero potential outliers in the examined numerical features.

Therefore:

No outlier removal or transformation was performed.

This decision avoided unnecessarily removing legitimate customer observations.

FEATURE SELECTION.

Feature selection was performed to remove irrelevant or highly redundant variables while retaining meaningful information.

- customerID

customerID was removed because it is an identifier and does not provide meaningful information about customer behavior.

- AverageMonthlySpend

AverageMonthlySpend was removed because its correlation with MonthlyCharges was approximately 0.99, indicating that it provided highly redundant information.

- Constant features

The dataset was checked for columns containing only one unique value.

The result was:

[]

Therefore, no constant features were removed.

- Correlated numerical features

tenure and TotalCharges showed a strong correlation of approximately 0.83. However, both were retained because they represent different business concepts:

tenure represents the length of the customer's relationship with the company.

TotalCharges represents the accumulated amount charged to the customer.

CATEGORICAL ENCODING.

The categorical variables were converted into numerical representations because machine-learning algorithms generally require numerical inputs.

One-hot encoding was applied to the categorical features using:

```
pd.get_dummies( X,  
                columns=categorical_features,  
                drop_first=True,  
                dtype=int)
```

drop_first=True was used to avoid unnecessary redundancy among dummy variables.

The target variable Churn was converted into binary values:

No → 0

Yes → 1

After encoding, the dataset was checked to ensure that no categorical object columns remained.

The result was:

```
Index([], dtype='object')
```

This confirmed that the categorical encoding was successful.

FEATURE SCALING.

Standardization was applied to the numerical features:

SeniorCitizen

tenure

MonthlyCharges

TotalCharges

TotalServices

StandardScaler was used to transform these variables so that they have approximately:

Mean = 0

Standard deviation = 1

The binary encoded categorical variables were retained as 0 and 1 rather than being standardized.

The scaling results confirmed that the numerical features had means approximately equal to zero and standard deviations approximately equal to one.

FINAL DATASET.

After preprocessing, the final dataset contained:

7,043 rows × 32 columns

The final dataset consisted entirely of numerical variables.

Its data types were:

float64 → 5 columns

int64 → 27 columns

There were:

0 missing values

0 duplicate rows

0 object/string columns

The final dataset therefore provides a suitable numerical representation for future machine-learning development.

FINAL PREPROCESSING SUMMARY.

Preprocessing Task

Action/Result

Dataset inspection

Completed

Missing-value handling

Completed

Duplicate check

0 duplicates

customerID

Removed

Feature engineering

TotalServices created

AverageMonthlySpend

Removed due to ~0.99 correlation with MonthlyCharges

Outlier detection

0 potential outliers detected

Constant-feature check

None found

Categorical encoding

One-hot encoding

Target encoding

No = 0, Yes = 1

Feature scaling

StandardScaler applied to numerical features

Final missing values

0

Final object columns

0

Final shape

7043 × 32

CONCLUSION.

The preprocessing stage successfully transformed the original Telco Customer Churn dataset into a clean and numerical machine-learning-ready dataset. Data-quality issues were investigated, irrelevant and redundant features were addressed, categorical variables were encoded, and numerical variables were standardized.

The resulting dataset contains 7,043 records and 32 columns, with no missing values, duplicate rows, or remaining categorical object columns.